الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

WEEK 9 · LESSONS 3-4 · 3-5

Weekly Quiz Preparation

Grade 12 Advanced · Mathematics

كتيّب التجهّز للاختبار الأسبوعي القصير

Mr. Yousef Al Yousef · **Friday 30 October 2026**

1 The product rule and tangent lines
قاعدة الضرب والمماسات

Differentiate a product with $(fg)' = f'g + fg'$. For a tangent line at $x = a$ find $h(a)$ and $h'(a)$, then write $y - h(a) = h'(a)(x - a)$.

مشتقة الضرب $(fg)' = f'g + fg'$. والمماس عند a: $y - h(a) = h'(a)(x - a)$ بعد حساب $h(a)$ و$h'(a)$.

Steps الخطوات

1	**Write $h = f \cdot g$ and find $h(a) = f(a)g(a)$**	احسب $h(a) = f(a)g(a)$
2	$h'(a) = f'(a)g(a) + f(a)g'(a)$	$h'(a) = f'g + fg'$
3	**Substitute the given values carefully**	عوّض القيم المعطاة
4	**$y - h(a) = h'(a)(x - a)$, then simplify**	اكتب معادلة المماس وبسّطها

Worked example مثال محلول

Assume $f(2) = 3$, $f'(2) = -1$ and $g(2) = 4$, $g'(2) = 2$. Find the tangent line to $y = h(x)$ at $x = 2$, where $h(x) = f(x)g(x)$.

$h(2) = f(2)g(2) = 3 \cdot 4 = 12$ — the point on the curve / النقطة على المنحنى

$h'(2) = f'(2)g(2) + f(2)g'(2) = (-1)(4) + (3)(2) = 2$ — the product rule / قاعدة الضرب

$y - 12 = 2(x - 2)$ — point-slope form / صيغة الميل ونقطة

$y = 2x + 8$ — simplify / بسّط

Watch out انتبه

- Both terms of the product rule: $f'g$ AND fg'. — حدّا قاعدة الضرب معاً: $f'g$ و fg'.
- The tangent line passes through (a, $h(a)$), not $(a, 0)$. — المماس يمرّ بالنقطة $(a, h(a))$ وليس $(a, 0)$.

Now you try جرّب بنفسك — solutions at the back الحلول في آخر الكتيّب

1. Find the tangent line to $f(x) = (x^3 + 2)(x - 1)$ at $x = 2$.
2. Assume $f(1) = 2$, $f'(1) = 3$ and $g(1) = -1$, $g'(1) = 4$. Find the tangent line to $h = fg$ at $x = 1$.
3. Find the tangent line to $f(x) = (x^2 - 3)(2x + 1)$ at $x = 1$.

From the End-of-Term structure. These are the exam exercises for this idea: practise solving them.

من هيكل نهاية الفصل: هذه تمارين الاختبار لهذه الفكرة، تدرّب على حلّها.

Question 11 | MCQ · on the computer | اختيار من متعدد · على الحاسوب

Exercises 21-24 · p. 169

In exercises 21–24, assume that f and g are differentiable with $f(0) = -1, f(1) = -2, f'(0) = -1, f'(1) = 3, g(0) = 3, g(1) = 1, g'(0) = -1$ and $g'(1) = -2$. Find an equation of the tangent line to the graph of $y = h(x)$ at $x = a$.

21. $h(x) = f(x)g(x)$; (a) $a = 0$; (b) $a = 1$

22. $h(x) = \dfrac{f(x)}{g(x)}$; (a) $a = 1$; (b) $a = 0$

23. $h(x) = x^2 f(x)$; (a) $a = 1$; (b) $a = 0$

24. $h(x) = \dfrac{x^2}{g(x)}$; (a) $a = 1$; (b) $a = 0$

2 The quotient rule and tangent lines
قاعدة القسمة والمماسات

Differentiate a quotient with $(\frac{f}{g})' = (f'g - fg')/g^2$. For a tangent line at $x = a$ use $h(a)$, $h'(a)$ and $y - h(a) = h'(a)(x - a)$.

مشتقة القسمة $(\frac{f}{g})' = (f'g - fg') \div g^2$. والمماس عند a: $y - h(a) = h'(a)(x - a)$.

Steps الخطوات

1	**Write $h = \frac{f}{g}$ and find $h(a) = f\frac{a}{g}(a)$**	احسب $h(a) = f\frac{a}{g}(a)$
2	**$h'(a) = (f'g - fg')/g^2$ at a**	$g^2/(f'g - fg') = h'(a)$
3	**Substitute the values, keep the order of the subtraction**	عوّض مع الحفاظ على ترتيب الطرح
4	**$y - h(a) = h'(a)(x - a)$, then simplify**	اكتب المماس وبسّطه

Worked example مثال محلول

Assume $f(0) = 2$, $f'(0) = -3$ and $g(0) = 1$, $g'(0) = 2$. Find the tangent line to $y = h(x)$ at $x = 0$, where $h(x) = \dfrac{f(x)}{g(x)}$.

$$h(0) = \frac{f(0)}{g(0)} = \frac{2}{1} = 2$$

the point on the curve — النقطة على المنحنى

$$h'(0) = \frac{f'(0)g(0) - f(0)g'(0)}{[g(0)]^2} = \frac{(-3)(1) - (2)(2)}{1} = -7$$

the quotient rule — قاعدة القسمة

$$y - 2 = -7(x - 0) \Rightarrow y = -7x + 2$$

point-slope form — صيغة الميل ونقطة

Watch out انتبه

- Divide by g SQUARED, not by g. — اقسم على g^2 وليس على g.
- The order is $f'g - fg'$; swapping them changes the sign. — الترتيب $f'g - fg'$؛ تبديلهما يغيّر الإشارة.

Now you try جرّب بنفسك — solutions at the back الحلول في آخر الكتيّب

1. Find the tangent line to $f(x) = \dfrac{x + 2}{x^2 + 3}$ at $x = 1$.
2. Assume $f(1) = 4$, $f'(1) = 2$ and $g(1) = 2$, $g'(1) = -1$. Find the tangent line to $h = \dfrac{f}{g}$ at $x = 1$.
3. Find the tangent line to $f(x) = \dfrac{3x - 1}{x + 1}$ at $x = 0$.

From the End-of-Term structure. These are the exam exercises for this idea: practise solving them.

من هيكل نهاية الفصل: هذه تمارين الاختبار لهذه الفكرة، تدرّب على حلّها.

Question 12 | MCQ · on the computer | اختيار من متعدد · على الحاسوب

Exercises 21-24 · p. 169

In exercises 21–24, assume that f and g are differentiable with $f(0) = -1, f(1) = -2, f'(0) = -1, f'(1) = 3, g(0) = 3, g(1) = 1, g'(0) = -1$ and $g'(1) = -2$. Find an equation of the tangent line to the graph of $y = h(x)$ at $x = a$.

21. $h(x) = f(x)g(x)$; (a) $a = 0$; (b) $a = 1$

22. $h(x) = \dfrac{f(x)}{g(x)}$; (a) $a = 1$; (b) $a = 0$

23. $h(x) = x^2 f(x)$; (a) $a = 1$; (b) $a = 0$

24. $h(x) = \dfrac{x^2}{g(x)}$; (a) $a = 1$; (b) $a = 0$

3 The chain rule
قاعدة السلسلة

For $h(x) = f(g(x))$: $h'(a) = f'(g(a)) \cdot g'(a)$. First find $g(a)$, then read f' at that value, then multiply by $g'(a)$.

للدالة $h(x) = f(g(x))$: المشتقة $h'(a) = f'(g(a)) \cdot g'(a)$. أوجد $g(a)$ أولاً ثم اقرأ f' عند هذه القيمة ثم اضرب في $g'(a)$.

Steps الخطوات

1	**Write $h'(a) = f'(g(a)) \cdot g'(a)$**	اكتب $h'(a) = f'(g(a)) \cdot g'(a)$
2	**Find the inside value $g(a)$**	أوجد القيمة الداخلية $g(a)$
3	**Read f' at $g(a)$, not at a**	اقرأ f' عند $g(a)$ لا عند a
4	**Multiply by $g'(a)$**	اضرب في $g'(a)$

Worked example مثال محلول

For $h(x) = f(g(x))$: $g(2) = 5$, $g'(2) = 3$, $f'(2) = 7$ and $f'(5) = -4$. Compute $h'(2)$.

$h'(2) = f'(g(2)) \cdot g'(2)$ — the chain rule / قاعدة السلسلة

$g(2) = 5$, so $f'(g(2)) = f'(5) = -4$ — the inside value first / القيمة الداخلية أولًا

$g'(2) = 3$ — the inside derivative / مشتقة الدالة الداخلية

$h'(2) = (-4)(3) = -12$ — multiply / اضرب

Watch out انتبه

- Read f' at $g(a)$, not at a. — اقرأ f' عند $g(a)$ وليس عند a.
- Do not forget the factor $g'(a)$. — لا تنسَ العامل $g'(a)$.

Now you try جرّب بنفسك — solutions at the back الحلول في آخر الكتيّب

1. For $h(x) = f(g(x))$: $g(1) = 4$, $g'(1) = -2$, $f'(1) = 3$ and $f'(4) = 5$. Compute $h'(1)$.
2. Let f be a differentiable function. Find $\frac{d}{dx}\left[f(x^3)\right]$.
3. Let f be a differentiable function. Find $\frac{d}{dx}\left[(f(x))^3\right]$.

From the End-of-Term structure. These are the exam exercises for this idea: practise solving them.

من هيكل نهاية الفصل: هذه تمارين الاختبار لهذه الفكرة، تدرّب على حلّها.

Question 13 | MCQ · on the computer | اختيار من متعدد · على الحاسوب

Exercises 31-36 · p. 177

In exercises 31 and 32, use the relevant information to compute the derivative for $h(x) = f(g(x))$.

31. $h'(1)$, where $f(1) = 3, g(1) = 2, f'(1) = 4, f'(2) = 3, g'(1) = -2$ and $g'(3) = 5$

32. $h'(2)$, where $f(2) = 1, g(2) = 3, f'(2) = -1, f'(3) = -3, g'(1) = 2$ and $g'(2) = 4$

. . . . . .

33. A function f is an **even function** if $f(-x) = f(x)$ for all x and is an **odd function** if $f(-x) = -f(x)$ for all x. Prove that the derivative of an even function is odd and the derivative of an odd function is even.

34. If the graph of a differentiable function f is symmetric about the line $x = a$, what can you say about the symmetry of the graph of f'?

In exercises 35–38, find the derivative where f is an unspecified differentiable function.

35. (a) $f(x^2)$ (b) $[f(x)]^2$ (c) $f(f(x))$

36. (a) $f(\sqrt{x})$ (b) $\sqrt{f(x)}$ (c) $f(xf(x))$

Solutions

written out step by step · الحلول

1 · The product rule and tangent lines — قاعدة الضرب والمماسات

1.1

Find the tangent line to $f(x) = (x^3 + 2)(x - 1)$ at $x = 2$.

$f(2) = (10)(1) = 10$ — the point on the curve · النقطة على المنحنى

$f'(x) = 3x^2(x-1) + (x^3+2)(1) \Rightarrow f'(2) = 12 + 10 = 22$ — the product rule · قاعدة الضرب

$y - 10 = 22(x - 2) \Rightarrow y = 22x - 34$ — point-slope form · صيغة الميل ونقطة

1.2

Assume $f(1) = 2$, $f'(1) = 3$ and $g(1) = -1$, $g'(1) = 4$. Find the tangent line to $h = fg$ at $x = 1$.

$h(1) = (2)(-1) = -2$ — the point on the curve · النقطة على المنحنى

$h'(1) = 3(-1) + (2)(4) = 5$ — the product rule · قاعدة الضرب

$y + 2 = 5(x - 1) \Rightarrow y = 5x - 7$ — point-slope form · صيغة الميل ونقطة

1.3

Find the tangent line to $f(x) = (x^2 - 3)(2x + 1)$ at $x = 1$.

$f(1) = (-2)(3) = -6$ — the point on the curve · النقطة على المنحنى

$f'(x) = 2x(2x+1) + (x^2 - 3)(2) \Rightarrow f'(1) = 6 - 4 = 2$ — the product rule · قاعدة الضرب

$y + 6 = 2(x - 1) \Rightarrow y = 2x - 8$ — point-slope form · صيغة الميل ونقطة

2 · The quotient rule and tangent lines — قاعدة القسمة والمماسات

2.1

Find the tangent line to $f(x) = \dfrac{x+2}{x^2+3}$ at $x = 1$.

$f(1) = \dfrac{3}{4}$ — the point on the curve · النقطة على المنحنى

$f'(x) = \dfrac{(1)(x^2+3) - (x+2)(2x)}{(x^2+3)^2} \Rightarrow f'(1) = \dfrac{4-6}{16} = -\dfrac{1}{8}$ — the quotient rule · قاعدة القسمة

$y - \frac{3}{4} = -\frac{1}{8}(x - 1) \Rightarrow y = -\frac{1}{8}x + \frac{7}{8}$ — point-slope form · صيغة الميل ونقطة

2.2

Assume $f(1)=4,\ f'(1)=2$ and $g(1)=2,\ g'(1)=-1$. Find the tangent line to $h=\frac{f}{g}$ at $x=1$.

$h(1)=\frac{4}{2}=2$ — the point on the curve / النقطة على المنحنى

$h'(1)=\frac{(2)(2)-(4)(-1)}{2^2}=\frac{8}{4}=2$ — the quotient rule / قاعدة القسمة

$y-2=2(x-1)\ \Rightarrow\ y=2x$ — point-slope form / صيغة الميل ونقطة

2.3

Find the tangent line to $f(x)=\frac{3x-1}{x+1}$ at $x=0$.

$f(0)=\frac{-1}{1}=-1$ — the point on the curve / النقطة على المنحنى

$f'(x)=\frac{3(x+1)-(3x-1)(1)}{(x+1)^2}=\frac{4}{(x+1)^2}\ \Rightarrow\ f'(0)=4$ — the quotient rule / قاعدة القسمة

$y+1=4x\ \Rightarrow\ y=4x-1$ — point-slope form / صيغة الميل ونقطة

3 · The chain rule — قاعدة السلسلة

3.1

For $h(x)=f(g(x))$: $g(1)=4$, $g'(1)=-2$, $f'(1)=3$ and $f'(4)=5$. Compute $h'(1)$.

$g(1)=4$, so $f'(g(1))=f'(4)=5$ — the inside value first, then f' there / القيمة الداخلية أولًا ثم f' عندها

$h'(1)=5\cdot(-2)=-10$ — multiply by $g'(1)$ / اضرب في $g'(1)$

3.2

Let f be a differentiable function. Find $\frac{d}{dx}\left[f(x^3)\right]$.

$\frac{d}{dx}f(x^3)=f'(x^3)\cdot\frac{d}{dx}x^3$ — the outer function is f, the inner function is x^3 / الدالة الخارجية f والدالة الداخلية x^3

$=3x^2f'(x^3)$ — the derivative of the inner function is $3x^2$ / مشتقة الدالة الداخلية $3x^2$

3.3

Let f be a differentiable function. Find $\frac{d}{dx}\left[(f(x))^3\right]$.

$\frac{d}{dx}(f(x))^3=3(f(x))^2\cdot f'(x)$ — the outer function is u^3, the inner function is $f(x)$ / الدالة الخارجية u^3 والدالة الداخلية $f(x)$

$=3(f(x))^2f'(x)$ — the derivative of the inner function is $f'(x)$ / مشتقة الدالة الداخلية $f'(x)$